

GARP Risk & AI — Q351–Q450

Unofficial Exam Prep

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Question

Demographic parity requires equality of:

- A. True positive rates
- B. False positive rates
- C. Positive prediction rates
- D. Precision

Answer: C

Explanation

Demographic parity requires:

$$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = b)$$

Positive prediction rates must be equal across groups.

Question

Equal opportunity requires equality of:

- A. Precision
- B. Sensitivity (True Positive Rate)
- C. Accuracy
- D. Positive prediction rate

Answer: B

Explanation

Equal opportunity requires:

$$P(\hat{Y} = 1 | Y = 1, A)$$

to be equal across groups.

Question

Predictive rate parity requires equality of:

- A. Precision
- B. Recall
- C. False Positive Rate
- D. Accuracy

Answer: A

Explanation

Predictive rate parity requires:

$$P(Y = 1 | \hat{Y} = 1, A)$$

to be equal across groups.

Question

Equalized odds requires equality of:

- A. Precision and Accuracy
- B. TPR and FPR
- C. Precision and Recall
- D. Positive Prediction Rate and Precision

Answer: B

Explanation

Equalized odds requires:

$$TPR_A = TPR_B$$

and

$$FPR_A = FPR_B$$

simultaneously.

Question

Which fairness metric conditions on predicted positives?

- A. Demographic Parity
- B. Equal Opportunity
- C. Predictive Rate Parity
- D. Equalized Odds

Answer: C

Explanation

Predictive rate parity evaluates:

$$P(Y = 1 | \hat{Y} = 1)$$

which conditions on predicted positives.

Question

Precision answers which question?

- A. Of all actual positives, how many were found?
- B. Of all predicted positives, how many were correct?
- C. Of all negatives, how many were correct?
- D. Of all predictions, how many were correct?

Answer: B

Explanation

$$\text{Precision} = \frac{TP}{TP + FP}$$

Question

Recall (Sensitivity) answers which question?

- A. Of all actual positives, how many were correctly identified?
- B. Of all predicted positives, how many were correct?
- C. Of all negatives, how many were correct?
- D. Of all predictions, how many were correct?

Answer: A

Explanation

$$\text{Recall} = \frac{TP}{TP + FN}$$

Question

False Positive Rate (FPR) measures:

- A. The fraction of actual positives missed
- B. The fraction of actual negatives incorrectly predicted positive
- C. The fraction of predicted positives that are wrong
- D. The fraction of predictions that are correct

Answer: B

Explanation

$$FPR = \frac{FP}{FP + TN}$$

Question

An ROC curve plots:

- A. Precision vs Recall
- B. Accuracy vs Threshold
- C. TPR vs FPR
- D. FPR vs Precision

Answer: C

Explanation

The ROC curve plots:

TPR

against

FPR

across thresholds.

Question

AUC is BEST interpreted as:

- A. Average precision
- B. Average accuracy
- C. Overall ability to distinguish positives from negatives
- D. Probability that a prediction is correct

Answer: C

Question

What AUC corresponds to a perfect classifier?

- A. 0
- B. 0.5
- C. 0.75
- D. 1.0

Answer: D

Question

An AUC of 0.5 suggests:

- A. Perfect classification
- B. Random guessing
- C. Severe overfitting
- D. High precision

Answer: B

Question

Lowering a decision threshold will generally:

- A. Increase Recall and Increase FPR
- B. Increase Recall and Decrease FPR
- C. Decrease Recall and Increase FPR
- D. Decrease Recall and Decrease FPR

Answer: A

Explanation

A lower threshold classifies more observations as positive, increasing both true positives and false positives.

Question

A fraud team lowers its model threshold. What is the MOST likely outcome?

- A. More fraud caught and more false alarms
- B. Less fraud caught and fewer false alarms
- C. Higher precision and lower recall
- D. Lower recall and higher specificity

Answer: A

Question

Predictive rate parity requires equality of:

- A. Positive prediction rates
- B. Precision across groups
- C. Sensitivity across groups
- D. False positive rates across groups

Answer: B

Precision across groups

Explanation

Predictive rate parity (also called conditional use accuracy equality or predictive parity) requires that the positive predictive value (precision) be equal across groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

In words: among all individuals who are *predicted* to be positive, the proportion who are *actually* positive must be the same for every group. This conditions on the prediction (the predicted positive class), not on the true label. It does *not* require equal positive prediction rates (that is demographic parity), nor equal sensitivity (that is equal opportunity), nor equal false positive rates (that is predictive equality).

Question

Equal opportunity requires equality of:

- A. Precision
- B. Overall accuracy
- C. Sensitivity (true positive rate)
- D. Positive prediction rate

Answer: C

Sensitivity (true positive rate)

Explanation

Equal opportunity requires that the true positive rate (sensitivity / recall) be equal across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

In words: among all individuals who are *truly* positive, the proportion who are *correctly* identified as positive must be the same for every group. This conditions on the true label (the actual positive class). It does *not* require equal precision (that is predictive rate parity), nor equal overall accuracy, nor equal positive prediction rates (that is demographic parity).

Question

Which fairness criterion conditions on the predicted positive class?

- A. Equal Opportunity
- B. Predictive Rate Parity
- C. Equalized Odds
- D. Demographic Parity

Answer: B

Predictive Rate Parity

Explanation

Predictive rate parity conditions on the predicted positive class ($\hat{Y} = 1$). It asks: given that the model predicted positive, what is the probability that the individual is truly positive?

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

This is exactly the definition of precision (positive predictive value). By contrast:

Equal Opportunity conditions on the *true* positive class ($Y = 1$) and equalizes sensitivity (TPR).

Equalized Odds conditions on *both* the true positive and true negative classes, equalizing TPR and FPR simultaneously.

Demographic Parity does not condition on any outcome label; it simply equalizes the positive prediction rate $P(\hat{Y} = 1 \mid A)$.

Question

Which fairness criterion conditions on those who are truly positive?

- A. Equal Opportunity
- B. Predictive Rate Parity
- C. Demographic Parity
- D. Predictive Equality

Answer: A

Equal Opportunity

Explanation

Equal opportunity conditions on the true positive class ($Y = 1$). It requires equal sensitivity (true positive rate) across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

In words: among those who are *truly* positive, the probability of being correctly predicted positive must be the same for all groups. Why the others are wrong:

Predictive Rate Parity conditions on the *predicted* positive class ($\hat{Y} = 1$), not the true positive class.

Demographic Parity does not condition on any outcome; it equalizes the positive prediction rate.

Predictive Equality conditions on the *true negative* class ($Y = 0$) and equalizes the false positive rate.

This question is a “reverse conditioning trap” because it swaps the conditioning direction relative to Q512.

Question

Demographic parity equalizes:

- A. Precision
- B. Sensitivity
- C. Positive prediction rates
- D. Accuracy

Answer: C

Positive prediction rates

Explanation

Demographic parity (also called statistical parity or independence) requires that the positive prediction rate be equal across groups:

$$P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$$

In words: the proportion of individuals predicted positive must be the same for every group, regardless of the true labels.

Why the others are wrong:

Precision is equalized by predictive rate parity.

Sensitivity (TPR) is equalized by equal opportunity.

Accuracy is not the target of any of these standard group-fairness criteria (though it may be equal in some settings).

Demographic parity is the only criterion here that can be evaluated *without* observing ground-truth outcomes, because it depends only on predictions and group membership.

Question

Which pairing is correct?

- A. Demographic Parity – Equal Precision
- B. Equal Opportunity – Equal Sensitivity
- C. Predictive Rate Parity – Equal Recall
- D. Predictive Equality – Equal Precision

Answer: B

Equal Opportunity – Equal Sensitivity

Explanation

The correct pairing is Equal Opportunity – Equal Sensitivity. Equal opportunity requires equal true positive rates (sensitivity/recall) across groups. Why the others are wrong:

- A: Demographic parity equalizes *positive prediction rates*, not precision.
- C: Predictive rate parity equalizes *precision* (positive predictive value), not recall (sensitivity).
- D: Predictive equality equalizes *false positive rates*, not precision.

A useful memory aid:

Criterion	Equalized Metric
Demographic Parity	Positive Prediction Rate
Equal Opportunity	True Positive Rate (Sensitivity/Recall)
Predictive Rate Parity	Precision (PPV)
Predictive Equality	False Positive Rate
Equalized Odds	TPR and FPR

Question

Predictive equality requires equality of:

- A. False positive rates
- B. Precision
- C. Accuracy
- D. Positive prediction rates

Answer: A

False positive rates

Explanation

Predictive equality requires that the false positive rate (FPR) be equal across groups:

$$P(\hat{Y} = 1 \mid Y = 0, A = 0) = P(\hat{Y} = 1 \mid Y = 0, A = 1)$$

In words: among all individuals who are *truly negative*, the proportion who are *incorrectly* predicted positive must be the same for every group. This conditions on the true negative class ($Y = 0$).

Why the others are wrong:

Precision is equalized by predictive rate parity.

Accuracy is not a group-fairness criterion in this family.

Positive prediction rates are equalized by demographic parity.

Predictive equality is sometimes called equal false positive rate and is one half of equalized odds (the other half being equal opportunity).

Question

Equalized odds requires equality of:

- A. Precision and accuracy
- B. Positive prediction rates and precision
- C. True positive rates and false positive rates
- D. Accuracy and sensitivity

Answer: C

True positive rates and false positive rates

Explanation

Equalized odds requires that both the true positive rate (TPR) and the false positive rate (FPR) be equal across groups:

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

$$P(\hat{Y} = 1 \mid Y = 0, A = 0) = P(\hat{Y} = 1 \mid Y = 0, A = 1)$$

In words: the model must have equal sensitivity *and* equal false positive rate for every group. Equivalently, the prediction $\hat{Y}$ must be independent of the protected attribute A *conditional on the true label* Y .

Why the others are wrong:

- A: Precision and accuracy are not the targets of equalized odds.
- B: Positive prediction rates and precision are targets of demographic parity and predictive rate parity, respectively.
- D: Accuracy and sensitivity are not jointly equalized by equalized odds (only TPR and FPR are).

Equalized odds is a combination of equal opportunity (equal TPR) and predictive equality (equal FPR).

Question

Which fairness criterion can be evaluated without observing ground-truth outcomes?

- A. Equal Opportunity
- B. Predictive Rate Parity
- C. Equalized Odds
- D. Demographic Parity

Answer: D

Demographic Parity

Explanation

Demographic parity can be evaluated without observing ground-truth outcomes because it depends only on the predicted labels and group membership:

$$P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$$

No true labels Y are needed.

Why the others require ground truth:

Equal Opportunity requires $Y = 1$ to compute sensitivity.

Predictive Rate Parity requires $Y = 1$ to compute precision.

Equalized Odds requires both $Y = 1$ and $Y = 0$ to compute TPR and FPR.

This is why demographic parity is often attractive in settings where labels are unavailable, delayed, or expensive to obtain. However, it can be satisfied even when the model is inaccurate, which is a known limitation.

Question

A candidate says: "Predictive rate parity equals the share of actual positives correctly identified."
Which metric are they accidentally describing?

- A. Predictive Rate Parity
- B. Demographic Parity
- C. Equal Opportunity
- D. Calibration

Answer: C

Equal Opportunity

Explanation

The phrase "the share of actual positives correctly identified" is the definition of sensitivity (true positive rate / recall):

$$\text{Sensitivity} = P(\hat{Y} = 1 \mid Y = 1)$$

Equal opportunity requires *equal* sensitivity across groups. So the candidate is accidentally describing Equal Opportunity, not predictive rate parity.
What predictive rate parity actually is: Predictive rate parity requires equal precision (positive predictive value) across groups:

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

Precision is "the share of *predicted* positives that are actually positive," which is the reverse conditioning of sensitivity.
Why the others are wrong:

- A: Predictive rate parity is precision, not sensitivity.
- B: Demographic parity is the positive prediction rate, $P(\hat{Y} = 1 \mid A)$.
- D: Calibration refers to the agreement between predicted probabilities and observed frequencies, not this quantity.

Key distinction:

Watch the video

Question

A bank wants to assess fairness immediately after making lending decisions. Loan defaults typically occur years later. Which fairness criterion is MOST practical at decision time?

- A. Predictive Parity
- B. Equal Opportunity
- C. Demographic Parity
- D. Equalized Odds

Answer: C

Explanation

Demographic parity does not require knowledge of future outcomes. The other metrics depend on ground-truth labels that become available only later.

Question

Why can demographic parity be problematic in medical diagnosis?

- A. It requires too much training data.
- B. It ignores legitimate differences in disease prevalence across groups.
- C. It requires causal models.
- D. It maximizes false negatives.

Answer: B

Explanation

If disease prevalence differs across groups, forcing equal positive prediction rates may reduce predictive usefulness and medical effectiveness.

Question

Which statement about demographic parity is TRUE?

- A. It guarantees equal error rates.
- B. It guarantees equal predictive value.
- C. It says nothing about whether predictions are correct.
- D. It guarantees equal opportunity.

Answer: C

Explanation

Demographic parity focuses only on equal prediction rates across groups and does not evaluate prediction quality.

Question

When base rates differ across groups, what does the fairness impossibility theorem imply?

- A. Demographic parity, predictive parity, and equal opportunity can generally be satisfied simultaneously.
- B. One fairness definition becomes objectively correct.
- C. Certain fairness criteria cannot all be satisfied at the same time.
- D. Fairness measurement becomes impossible.

Answer: C

Explanation

The theorem highlights that competing fairness definitions may conflict when underlying group base rates differ.

Question

A fraud model is deployed in an environment where confirmed fraud labels may take months to establish and many fraud cases are never confirmed. Why might demographic parity still be considered?

- A. It guarantees fairness.
- B. It does not require reliable ground-truth labels.
- C. It guarantees equal precision.
- D. It eliminates bias.

Answer: B

Explanation

Demographic parity can be evaluated without waiting for true fraud outcomes, making it attractive in settings where labels are delayed, incomplete, or extremely rare.